

EA-0061 - AQELYN Engineering Archive

EA-0061 — Developer Handbook & Implementation Guide

****Brand:**** AQELYN

****Subtitle:**** Practical guide for Codex, engineers, and maintainers implementing AQELYN

****Status:**** FULL_COMPLETE

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Executive Summary

Provide the implementation playbook from local setup through C-001 Foundation Runtime and the full module-by-module development roadmap.

This document is part of the AQELYN / AQELYN Architecture Baseline v1.0 and is normative for implementation. It supports the product principles: Explain Before You Recommend, Simplicity First, Evidence Before Opinion, Human-Centered Security, and Expert Depth on Demand.

Scope Domains

- Local environment setup
- Containerized development
- Codex workflow
- C-001 Foundation Runtime
- Module implementation sequence
- Testing workflow
- Debugging and observability
- Deployment and operations

01. Developer Onboarding

This section defines the mandatory rules, constraints, and implementation expectations for ****Developer Onboarding**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0061-REQ-01-01:**** The implementation shall apply developer onboarding consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-01-02:**** The implementation shall apply developer onboarding consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-01-03:**** The implementation shall apply developer onboarding consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-01-04:**** The implementation shall apply developer onboarding consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-01-05:**** The implementation shall apply developer onboarding consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Developer Onboarding is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Developer Onboarding is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Developer Onboarding is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Developer Onboarding is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Developer Onboarding, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

02. Local Setup

This section defines the mandatory rules, constraints, and implementation expectations for **Local Setup** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-02-01:** The implementation shall apply local setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-02-02:** The implementation shall apply local setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-02-03:** The implementation shall apply local setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-02-04:** The implementation shall apply local setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-02-05:** The implementation shall apply local setup consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Local Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Local Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Local Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Local Setup is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Local Setup, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

03. Toolchain

This section defines the mandatory rules, constraints, and implementation expectations for **Toolchain** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-03-01**: The implementation shall apply toolchain consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-03-02**: The implementation shall apply toolchain consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-03-03**: The implementation shall apply toolchain consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-03-04**: The implementation shall apply toolchain consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-03-05**: The implementation shall apply toolchain consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Toolchain is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Toolchain is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Toolchain is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Toolchain is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Toolchain, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

04. Environment Variables

This section defines the mandatory rules, constraints, and implementation expectations for **Environment Variables** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-04-01:** The implementation shall apply environment variables consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-04-02:** The implementation shall apply environment variables consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-04-03:** The implementation shall apply environment variables consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-04-04:** The implementation shall apply environment variables consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-04-05:** The implementation shall apply environment variables consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Environment Variables is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Environment Variables is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Environment Variables is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Environment Variables is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Environment Variables, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

05. Docker Stack

This section defines the mandatory rules, constraints, and implementation expectations for **Docker Stack** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-05-01:** The implementation shall apply docker stack consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-05-02:** The implementation shall apply docker stack consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-05-03:** The implementation shall apply docker stack consistently across all affected modules and maintain traceability to source requirements.

- ****EA-0061-REQ-05-04:**** The implementation shall apply docker stack consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-05-05:**** The implementation shall apply docker stack consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Docker Stack is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Docker Stack is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Docker Stack is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Docker Stack is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Docker Stack, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

06. Database Setup

This section defines the mandatory rules, constraints, and implementation expectations for ****Database Setup**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0061-REQ-06-01:**** The implementation shall apply database setup consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-06-02:**** The implementation shall apply database setup consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-06-03:**** The implementation shall apply database setup consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-06-04:**** The implementation shall apply database setup consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-06-05:**** The implementation shall apply database setup consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.

- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Database Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Database Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Database Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Database Setup is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Database Setup, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

07. Frontend Setup

This section defines the mandatory rules, constraints, and implementation expectations for **Frontend Setup** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-07-01:** The implementation shall apply frontend setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-07-02:** The implementation shall apply frontend setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-07-03:** The implementation shall apply frontend setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-07-04:** The implementation shall apply frontend setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-07-05:** The implementation shall apply frontend setup consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Frontend Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Frontend Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Frontend Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Frontend Setup is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Frontend Setup, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

08. Backend Setup

This section defines the mandatory rules, constraints, and implementation expectations for **Backend Setup** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-08-01:** The implementation shall apply backend setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-08-02:** The implementation shall apply backend setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-08-03:** The implementation shall apply backend setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-08-04:** The implementation shall apply backend setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-08-05:** The implementation shall apply backend setup consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Backend Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Backend Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Backend Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Backend Setup is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Backend Setup, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

09. Codex Workflow

This section defines the mandatory rules, constraints, and implementation expectations for **Codex Workflow** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-09-01:** The implementation shall apply codex workflow consistently across all affected modules and maintain traceability to source requirements.

- ****EA-0061-REQ-09-02:**** The implementation shall apply codex workflow consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-09-03:**** The implementation shall apply codex workflow consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-09-04:**** The implementation shall apply codex workflow consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-09-05:**** The implementation shall apply codex workflow consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Codex Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Codex Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Codex Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Codex Workflow is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Codex Workflow, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

10. C-001 Foundation Runtime

This section defines the mandatory rules, constraints, and implementation expectations for ****C-001 Foundation Runtime**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0061-REQ-10-01:**** The implementation shall apply c-001 foundation runtime consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-10-02:**** The implementation shall apply c-001 foundation runtime consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-10-03:**** The implementation shall apply c-001 foundation runtime consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-10-04:**** The implementation shall apply c-001 foundation runtime consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-10-05:**** The implementation shall apply c-001 foundation runtime consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: C-001 Foundation Runtime is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: C-001 Foundation Runtime is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: C-001 Foundation Runtime is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: C-001 Foundation Runtime is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to C-001 Foundation Runtime, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

11. Module Coding Order

This section defines the mandatory rules, constraints, and implementation expectations for **Module Coding Order** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-11-01:** The implementation shall apply module coding order consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-11-02:** The implementation shall apply module coding order consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-11-03:** The implementation shall apply module coding order consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-11-04:** The implementation shall apply module coding order consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-11-05:** The implementation shall apply module coding order consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Module Coding Order is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Module Coding Order is implemented, tested, documented, and linked to the traceability matrix.

- Acceptance item 3: Module Coding Order is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Module Coding Order is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Module Coding Order, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

12. Testing Workflow

This section defines the mandatory rules, constraints, and implementation expectations for **Testing Workflow** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-12-01:** The implementation shall apply testing workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-12-02:** The implementation shall apply testing workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-12-03:** The implementation shall apply testing workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-12-04:** The implementation shall apply testing workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-12-05:** The implementation shall apply testing workflow consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Testing Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Testing Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Testing Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Testing Workflow is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Testing Workflow, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

13. Debugging Guide

This section defines the mandatory rules, constraints, and implementation expectations for **Debugging Guide** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-13-01:** The implementation shall apply debugging guide consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-13-02:** The implementation shall apply debugging guide consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-13-03:** The implementation shall apply debugging guide consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-13-04:** The implementation shall apply debugging guide consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-13-05:** The implementation shall apply debugging guide consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Debugging Guide is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Debugging Guide is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Debugging Guide is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Debugging Guide is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Debugging Guide, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

14. Observability Setup

This section defines the mandatory rules, constraints, and implementation expectations for **Observability Setup** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-14-01:** The implementation shall apply observability setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-14-02:** The implementation shall apply observability setup consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-14-03:** The implementation shall apply observability setup consistently across all affected modules and maintain traceability to source requirements.

- ****EA-0061-REQ-14-04:**** The implementation shall apply observability setup consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-14-05:**** The implementation shall apply observability setup consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Observability Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Observability Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Observability Setup is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Observability Setup is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Observability Setup, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

15. Security Checklist

This section defines the mandatory rules, constraints, and implementation expectations for ****Security Checklist**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0061-REQ-15-01:**** The implementation shall apply security checklist consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-15-02:**** The implementation shall apply security checklist consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-15-03:**** The implementation shall apply security checklist consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-15-04:**** The implementation shall apply security checklist consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-15-05:**** The implementation shall apply security checklist consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.

- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Security Checklist is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Security Checklist is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Security Checklist is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Security Checklist is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Security Checklist, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

16. Release Workflow

This section defines the mandatory rules, constraints, and implementation expectations for **Release Workflow** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-16-01:** The implementation shall apply release workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-16-02:** The implementation shall apply release workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-16-03:** The implementation shall apply release workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-16-04:** The implementation shall apply release workflow consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-16-05:** The implementation shall apply release workflow consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Release Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Release Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Release Workflow is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Release Workflow is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Release Workflow, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

17. Deployment Guide

This section defines the mandatory rules, constraints, and implementation expectations for **Deployment Guide** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-17-01:** The implementation shall apply deployment guide consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-17-02:** The implementation shall apply deployment guide consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-17-03:** The implementation shall apply deployment guide consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-17-04:** The implementation shall apply deployment guide consistently across all affected modules and maintain traceability to source requirements.
- **EA-0061-REQ-17-05:** The implementation shall apply deployment guide consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Deployment Guide is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Deployment Guide is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Deployment Guide is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Deployment Guide is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Deployment Guide, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

18. Operational Runbooks

This section defines the mandatory rules, constraints, and implementation expectations for **Operational Runbooks** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- **EA-0061-REQ-18-01:** The implementation shall apply operational runbooks consistently across all affected modules and maintain traceability to source requirements.

- ****EA-0061-REQ-18-02:**** The implementation shall apply operational runbooks consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-18-03:**** The implementation shall apply operational runbooks consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-18-04:**** The implementation shall apply operational runbooks consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-18-05:**** The implementation shall apply operational runbooks consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Operational Runbooks is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Operational Runbooks is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Operational Runbooks is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Operational Runbooks is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Operational Runbooks, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

19. Troubleshooting

This section defines the mandatory rules, constraints, and implementation expectations for ****Troubleshooting**** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0061-REQ-19-01:**** The implementation shall apply troubleshooting consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-19-02:**** The implementation shall apply troubleshooting consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-19-03:**** The implementation shall apply troubleshooting consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-19-04:**** The implementation shall apply troubleshooting consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-19-05:**** The implementation shall apply troubleshooting consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Troubleshooting is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Troubleshooting is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 3: Troubleshooting is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Troubleshooting is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Troubleshooting, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

20. Developer FAQ

This section defines the mandatory rules, constraints, and implementation expectations for **Developer FAQ** within the AQELYN Platform. The guidance is designed to be directly actionable by Codex and human engineers.

Requirements

- ****EA-0061-REQ-20-01:**** The implementation shall apply developer faq consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-20-02:**** The implementation shall apply developer faq consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-20-03:**** The implementation shall apply developer faq consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-20-04:**** The implementation shall apply developer faq consistently across all affected modules and maintain traceability to source requirements.
- ****EA-0061-REQ-20-05:**** The implementation shall apply developer faq consistently across all affected modules and maintain traceability to source requirements.

Implementation Rules

- All code and configuration must be deterministic, reviewable, and testable.
- All external inputs must be validated before processing.
- All user-facing outputs must be understandable by non-experts and expandable for experts.
- All security decisions must include evidence references and audit metadata.
- All deviations require an Architecture Decision Record and engineering review.

Acceptance Criteria

- Acceptance item 1: Developer FAQ is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 2: Developer FAQ is implemented, tested, documented, and linked to the traceability matrix.

- Acceptance item 3: Developer FAQ is implemented, tested, documented, and linked to the traceability matrix.
- Acceptance item 4: Developer FAQ is implemented, tested, documented, and linked to the traceability matrix.

Codex Implementation Notes

Codex shall treat this section as binding. When implementing features related to Developer FAQ, it shall generate unit tests, integration tests where applicable, and update documentation artifacts in the same pull request.

Requirements Matrix

Requirement ID	Requirement	Priority	Verification
EA-0061-REQ-01	Implement Developer Onboarding controls	Mandatory	Review + Tests
EA-0061-REQ-02	Implement Local Setup controls	Mandatory	Review + Tests
EA-0061-REQ-03	Implement Toolchain controls	Mandatory	Review + Tests
EA-0061-REQ-04	Implement Environment Variables controls	Mandatory	Review + Tests
EA-0061-REQ-05	Implement Docker Stack controls	Mandatory	Review + Tests
EA-0061-REQ-06	Implement Database Setup controls	Mandatory	Review + Tests
EA-0061-REQ-07	Implement Frontend Setup controls	Mandatory	Review + Tests
EA-0061-REQ-08	Implement Backend Setup controls	Mandatory	Review + Tests
EA-0061-REQ-09	Implement Codex Workflow controls	Mandatory	Review + Tests
EA-0061-REQ-10	Implement C-001 Foundation Runtime controls	Mandatory	Review + Tests
EA-0061-REQ-11	Implement Module Coding Order controls	Mandatory	Review + Tests
EA-0061-REQ-12	Implement Testing Workflow controls	Mandatory	Review + Tests
EA-0061-REQ-13	Implement Debugging Guide controls	Mandatory	Review + Tests
EA-0061-REQ-14	Implement Observability Setup controls	Mandatory	Review + Tests
EA-0061-REQ-15	Implement Security Checklist controls	Mandatory	Review + Tests
EA-0061-REQ-16	Implement Release Workflow controls	Mandatory	Review + Tests
EA-0061-REQ-17	Implement Deployment Guide controls	Mandatory	Review + Tests
EA-0061-REQ-18	Implement Operational Runbooks controls	Mandatory	Review + Tests
EA-0061-REQ-19	Implement Troubleshooting controls	Mandatory	Review + Tests
EA-0061-REQ-20	Implement Developer FAQ controls	Mandatory	Review + Tests

Traceability Matrix

Requirement	Source	Implementation Target	Verification Evidence
EA-0061-REQ-01	EA-0061 Section 01	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-02	EA-0061 Section 02	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-03	EA-0061 Section 03	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-04	EA-0061 Section 04	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-05	EA-0061 Section 05	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-06	EA-0061 Section 06	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-07	EA-0061 Section 07	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-08	EA-0061 Section 08	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-09	EA-0061 Section 09	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-10	EA-0061 Section 10	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-11	EA-0061 Section 11	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist

EA-0061-REQ-12	EA-0061 Section 12	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-13	EA-0061 Section 13	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-14	EA-0061 Section 14	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-15	EA-0061 Section 15	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-16	EA-0061 Section 16	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-17	EA-0061 Section 17	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-18	EA-0061 Section 18	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-19	EA-0061 Section 19	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist
EA-0061-REQ-20	EA-0061 Section 20	src/, tests/, api/, sdk/ as applicable	CI logs, unit tests, review checklist

Engineering Journal

- 2026-07-09: EA-0061 created as part of the final architecture completion set EA-0058 through EA-0061.
- Decision: AQELYN branding is applied while AQELYN remains the internal engineering codename.
- Decision: this package is normative for coding readiness and Codex execution.

Final Review

The package is approved for implementation baseline use when all files are present, hashes are generated, diagrams render, and PDF/HTML outputs correspond to the Master Markdown.